

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics

PRELIMINARY HSC ASSESSMENT TASK JULY 2006

General Instructions

- Working time allowed – 70 minutes.
- Write using black or blue pen.
- Approved calculators may be used.
- All necessary working should be shown.
- Start each question on a new page.
- Attempt all questions.
- Questions are not of equal value.
- Full marks may not be awarded if working is poorly set out or difficult to read.

NAME : _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	TOTAL

Question 1 (7 marks)**Marks**

- a) The line $ax + 2y - 10 = 0$ passes through the point $(4, -1)$.

2

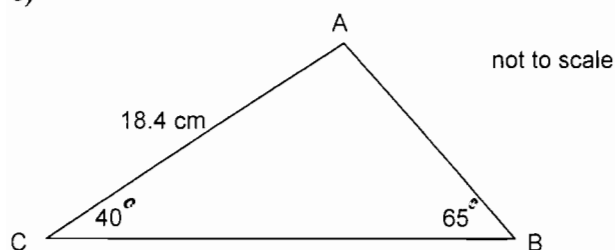
Find the value of a .

- b) Find the exact value of
- i) $\tan 30^\circ$
 - ii) $\sec 150^\circ$

1

1

c)



- i) Find the length of side AB correct to the nearest millimetre.
- ii) Find the area of triangle ABC correct to the nearest square centimetre.

2

1

Question 2 (7 marks) (Start a new page)

- a) Sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.

2

- b) Find the equation of the line perpendicular to $y = 2x - 9$ which passes through the point $(0, 3)$. Express your answer in general form.

3

- c) Find the acute angle that the line $3x - y - 5 = 0$ makes with the x axis.
(nearest degree)

2

Question 3 (7 marks) (Start a new page)

- a) Simplify $\cos \theta \tan \theta$. 1
- b) Solve $3 \tan \theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$ 2
Give answers correct to the nearest degree.
- c) Find the point of intersection of the lines 2
 $x - 2y - 7 = 0$ and $3x - 4y - 19 = 0$.
- d) Sketch the region $x + 2y + 2 \geq 0$. 2

Question 4 (8 marks) (Start a new page)

The points A , B and C have coordinates $(-3,0)$, $(3,4)$ and $(5,1)$ respectively.

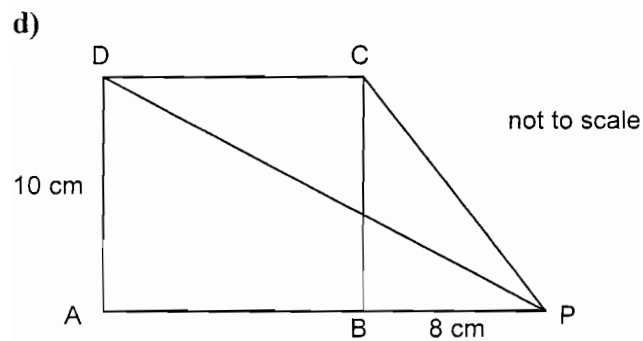
- a) Draw a neat number plane showing this information. 1
- b) Find the coordinates of the midpoint of AB. 1
- c) Find the gradient of AB. 1
- d) Find the equation of the line AB. 1
- e) Find the length of AB. (exact form) 1
- f) Find the equation of the circle with diameter AB. 2
- g) Find the coordinates of the point D such that ABCD is a parallelogram. 1

Question 5 (8 marks) (Start a new page)

a) Given $\sin \theta = \frac{3}{4}$ and $\cos \theta < 0$ find the exact value of $\tan \theta$. 2

b) Find the perpendicular distance from the point $(-1, 6)$ 2
to the line $x + 2y - 3 = 0$.

c) Find a value of θ so that $\sin(\theta + 30)^\circ = \cos 50^\circ$. 1



ABCD is a square with sides 10cm.

AB is extended to P such that BP=8cm.

Find $\angle DPC$ (correct to the nearest degree). 3

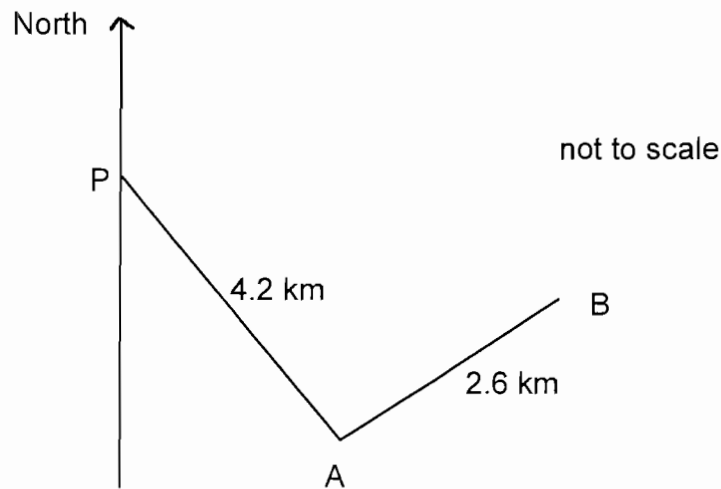
Question 6 (7 marks) (Start a new page)

a) If the lines $5x - 2y + 4 = 0$ and $y = mx - 2$ are parallel,

2

find the value of m

b)



Jay and Rob sail a boat from port P on a bearing of 165° for 4.2 km to position A.

The boat then sails on a bearing of 070° for 2.6 km to point B.

i) Find the size of angle PAB.

1

ii) Use the cosine rule to find the distance of position B from P.
Give answer in kilometres correct to 1 decimal place.

2

iii) Find the bearing of the port P from point B.
Give answer correct to the nearest degree.

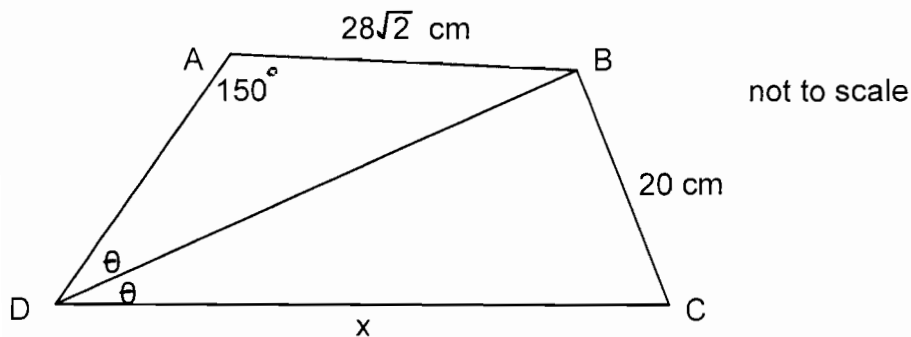
2

Question 7 (8 marks) (Start a new page)

a) Show that $(1 - \cos A)(1 + \sec A) = \sin A \tan A$

3

b)



In the above quadrilateral ABCD the diagonal BD bisects angle ADC and $BD = DC$.

$AB = 28\sqrt{2} \text{ cm}$, $BC = 20 \text{ cm}$ and $\angle DAB = 150^\circ$.

Let $\angle BDC = \theta$ and $DC = x \text{ cm}$

i) Using the sine rule in triangle ABD, show that $\sin \theta = \frac{14\sqrt{2}}{x}$.

2

ii) By considering triangle BDC, use the cosine rule to show that

1

$$\cos \theta = \frac{x^2 - 200}{x^2}$$

iii) Use the results in parts i) and ii) to find the exact value of x .

2

2 UNIT JULY SOLUTIONS YR 11

Question 1

a. $4a - 2 - 10 = 0$

$$4a = 12$$

$$a = 3$$

b. i) $\frac{1}{\sqrt{3}}$

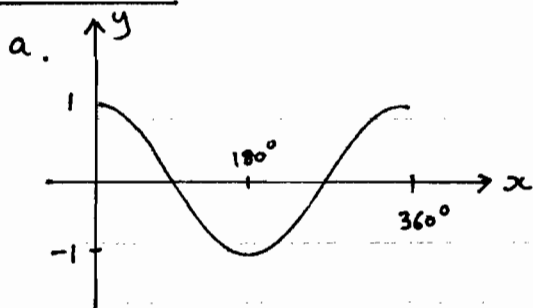
ii) $-\frac{2}{\sqrt{3}}$

c. i) $\frac{AB}{\sin 40^\circ} = \frac{18.4}{\sin 65^\circ}$

$$AB = 13.0 \text{ cm}$$

ii) $\text{Area} = \frac{1}{2} \times 18.4 \times 13.0 \times \sin 75^\circ$
 $= 116 \text{ cm}^2$

Question 2



b. $m = -\frac{1}{2} \quad (0, 3)$

$$\therefore y = -\frac{1}{2}x + 3$$

$$2y = -x + 6$$

$$x + 2y - 6 = 0$$

c. $m = 3$

$$\therefore \tan \theta = 3$$

$$\theta = 72^\circ$$

Question 3

a. $\cos \theta \tan \theta$

$$= \cos \theta \frac{\sin \theta}{\cos \theta}$$

$$= \sin \theta$$

b. $3 \tan \theta = 5$

$$\tan \theta = \frac{5}{3}$$

(1st, 3rd quad)

$$\theta = 59^\circ, 239^\circ$$

c. $x - 2y - 7 = 0$

$$3x - 4y - 19 = 0$$

$$2x - 4y - 14 = 0$$

$$3x - 4y - 19 = 0$$

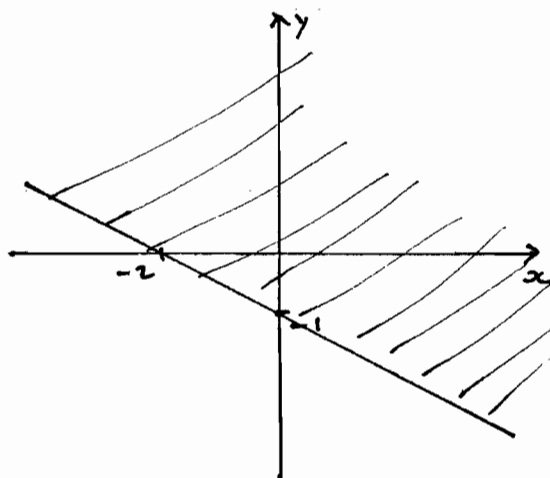
$$x - 5 = 0$$

$$x = 5$$

$$\therefore y = -1$$

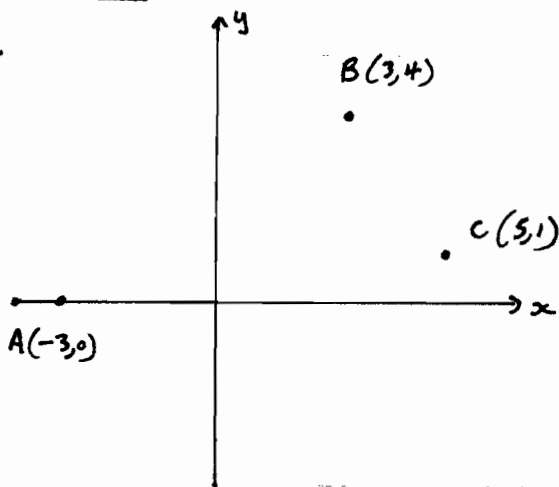
$\therefore$ pt. of intersection $(5, -1)$

d.



Question 4

a.



b. midpoint = $(0, 2)$

c. $m = \frac{4-0}{3-(-3)}$
 $= \frac{4}{6}$
 $= \frac{2}{3}$

d. $y - 4 = \frac{2}{3}(x - 3)$
 $3y - 12 = 2x - 6$
 $2x - 3y + 6 = 0$

e. $d = \sqrt{(3 - (-3))^2 + (4 - 0)^2}$
 $= \sqrt{52}$ units

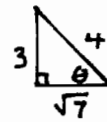
f. radius = $\frac{\sqrt{52}}{2}$
 $= \sqrt{13}$
 centre = $(0, 2)$

$\therefore$ equation $x^2 + (y - 2)^2 = 13$

g. $D(-1, -3)$

Question 5

a. 2nd quadrant



$\therefore \tan \theta = -\frac{3}{\sqrt{7}}$

b. $d = \frac{|-1 + 12 - 3|}{\sqrt{1^2 + 2^2}}$
 $= \frac{8}{\sqrt{5}}$ units

c. $\theta + 30 + 50 = 90$
 $\theta = 10$

d. $\tan \angle DPA = \frac{10}{18}$
 $\therefore \angle DPA = 29.05^\circ$
 $\tan \angle CPA = \frac{10}{8}$
 $\therefore \angle CPA = 51.34^\circ$
 $\therefore \angle DPC = 22^\circ$

Question 6

a. $5x - 2y + 4 = 0$

has gradient $\frac{5}{2}$

$\therefore m = \frac{5}{2}$

b. i) $\angle PAB = 85^\circ$

ii) $d^2 = 4 \cdot 2^2 + 2 \cdot 6^2 - 2 \times 4 \cdot 2 \times 2 \cdot 6 \times \cos 85^\circ$

$\therefore d = 4.7 \text{ km}$

iii) $\frac{\sin \angle PBA}{4 \cdot 2} = \frac{\sin 85^\circ}{4 \cdot 7}$

$\therefore \angle PBA = 63^\circ$

$\therefore \text{bearing} = 180^\circ + 70^\circ + 63^\circ$
 $= 313^\circ$

b. i)

$$\frac{\sin \theta}{28\sqrt{2}} = \frac{\sin 150^\circ}{x}$$

$$\sin \theta = \frac{28\sqrt{2} \sin 150^\circ}{x}$$

$$= \frac{28\sqrt{2} \times \frac{1}{2}}{x}$$

$$= \frac{14\sqrt{2}}{x}$$

ii)

$$\cos \theta = \frac{x^2 + x^2 - 20^2}{2x^2}$$

$$= \frac{2x^2 - 400}{2x^2}$$

$$= \frac{x^2 - 200}{x^2}$$

iii)

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\therefore \left(\frac{14\sqrt{2}}{x}\right)^2 + \left(\frac{x^2 - 200}{x^2}\right)^2 = 1$$

$$\frac{392}{x^2} + \frac{x^4 - 400x^2 + 40000}{x^4} = 1$$

$$392x^2 + x^4 - 400x^2 + 40000 = x^4$$

$$8x^2 = 40000$$

$$x^2 = 5000$$

$$x = 50\sqrt{2}$$

Question 7

a. LHS = $(1 - \cos A)(1 + \sec A)$

$$= 1 + \sec A - \cos A - \cos A \sec A$$

$$= \sec A - \cos A$$

$$= \frac{1 - \cos^2 A}{\cos A}$$

$$= \frac{\sin^2 A}{\cos A}$$

$$= \sin A \tan A$$

$$= \text{RHS}$$